



IndianOil

JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal to THIRUPPATHI VENGATESA AGENCIES

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable

recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

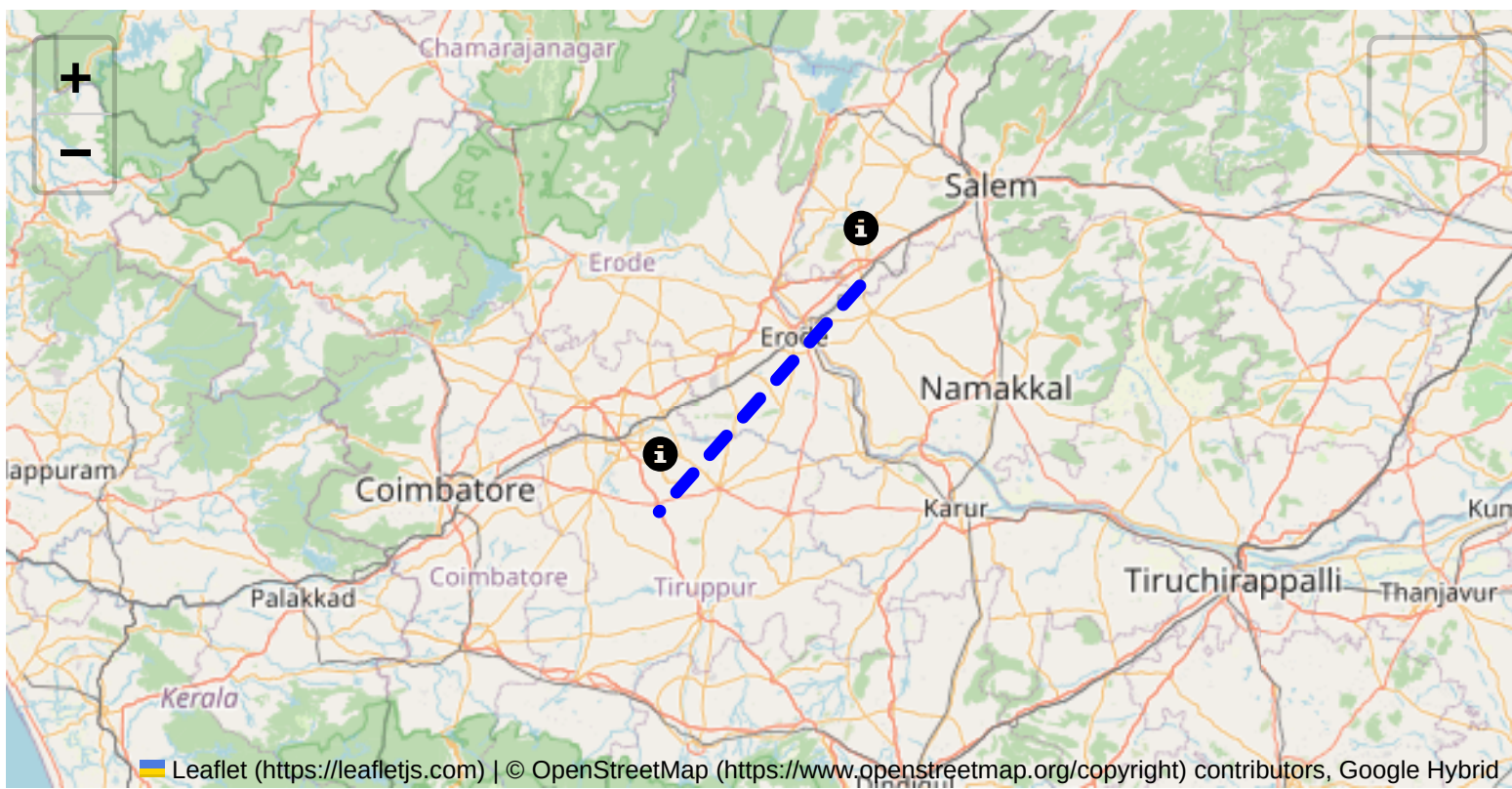
The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

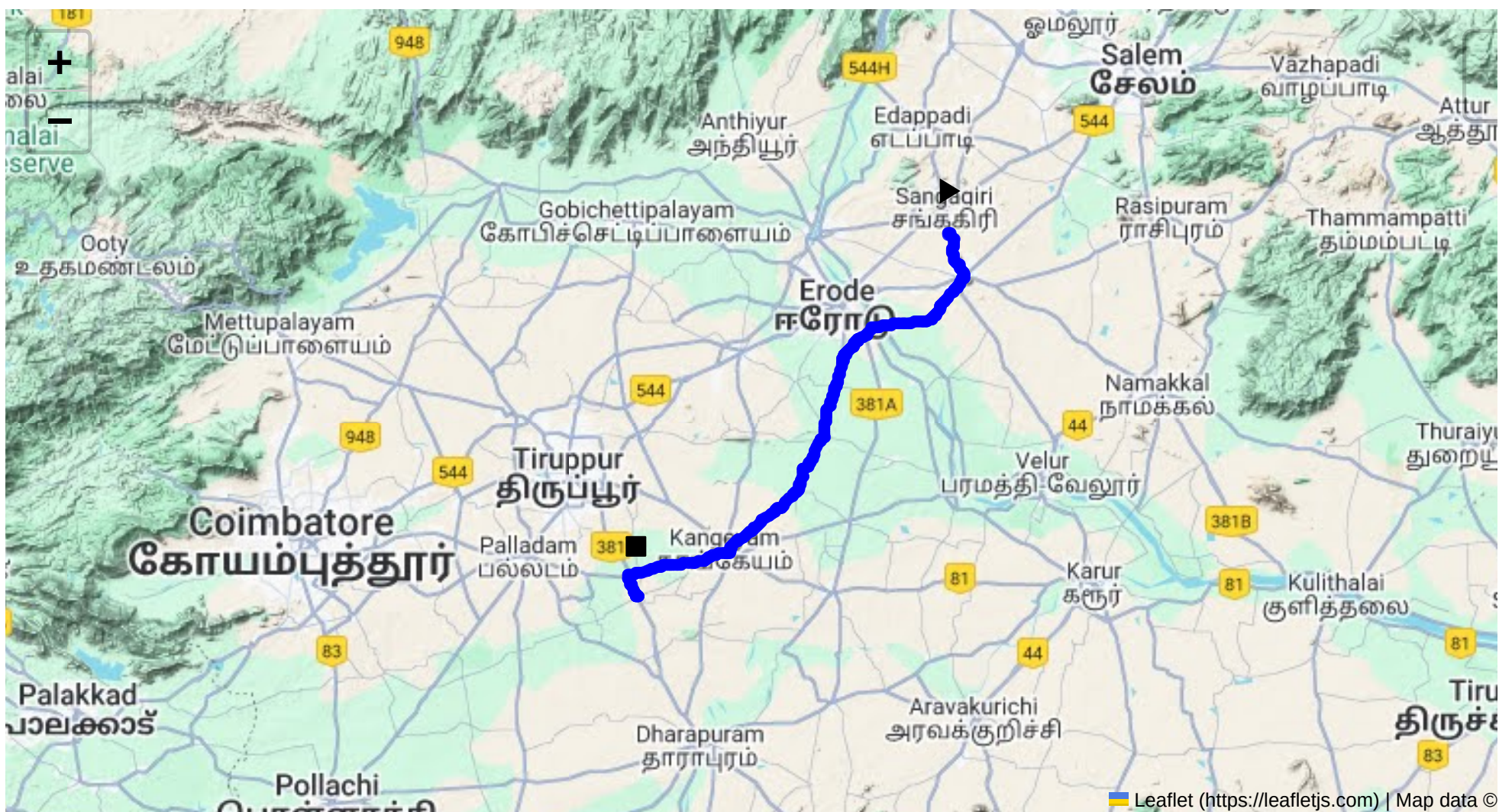
The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



Route Summary:
Total Distance: 90.74 km
Estimated Duration: 2.1 hours
Adjusted Duration (Heavy Vehicle): 2.6 hours
Start: (11.4381, 77.8734)
End: (10.949872, 77.433807)



Welcome to the Journey Risk Management Study

Route Analysis Overview

1. Overview of the Route Map:

- The route from CVQF+23W, Sangagiri to WCXM+XG9, South Avanashipalayam covers approximately 90.74 kilometers. The journey primarily takes place on the NH 544 highway followed by a transition to local roads. NH 544 is a major highway connecting various towns and

is generally well-maintained. The route passes through a mix of urban, semi-urban, and rural areas, providing varied terrain and traffic conditions.

2. Typical Weather Conditions and Potential Weather-Related Hazards:

- Tamil Nadu experiences a tropical climate with monsoon seasons from June to September and from October to December. Heavy rainfall during these periods can lead to waterlogged roads, reduced visibility, and potential flooding, especially in low-lying areas. High temperatures are common during summer months, which could lead to road deformations and affect vehicle performance.

3. Traffic Patterns:

- Traffic is generally denser during morning (7:30 - 9:30 AM) and evening (5:30 - 7:30 PM) rush hours, especially near urban centers like Erode. Congestion is common at intersections and toll booths on NH 544. Major cities and towns can cause slowdowns, with potential bottlenecks at market areas and bus stops.

4. Assessment of Road Quality and Infrastructure:

- NH 544 is a national highway with good road quality, multiple lanes, and regular maintenance. However, local roads toward South Avanashipalayam might be narrower with variable surface conditions and less lighting which can pose risks during nighttime driving.

5. Suggestions for Alternative Routes in Emergencies:

- Alternative roads include using SH 81 or regional roads that parallel NH 544. During emergencies, consider major district roads that bypass town centers and congested areas. Locally known bypasses can also help avoid traffic snarls or roadwork zones.

6. Summary of Local Regulations Affecting Hazardous Material Transport:

- Transporting hazardous materials requires adherence to specified guidelines like having proper permits, using marked vehicles, and following designated routes. Local regulations may restrict entry into specific city areas during certain times to avoid congestion.

7. Overview of Historical Incidents:

- While specific historical incident data may not be publicly detailed, NH 544 has been the site of various traffic accidents, including heavy vehicles. Safety measures have been heightened on this route with increased patrols and monitoring during peak seasons.

8. Environmental Considerations and Sensitive Areas:

- The route passes near agricultural zones and small water bodies. Special care should be taken to prevent contamination from spills. Avoid routes close to wildlife reserves or conservation areas to minimize ecological disturbance.

9. Analysis of Communication Coverage:

- Major parts of the route, especially on NH 544, have good network coverage from major telecom providers. Some rural parts might have weaker signals or dead zones impacting communication, particularly beyond urban areas.

10. Estimated Emergency Response Times:

- Emergency services have reasonably quick access to the highway due to its connectivity and police patrol presence. However, response times may increase in rural sections of the route or during adverse weather conditions and heavy traffic.

11. Overall Summary of Risk Assessment:

- Overall, the route presents moderate risks predominantly due to traffic congestion and variable weather conditions. The well-maintained highways reduce the risk associated with road infrastructure, although attention is needed for emergency scenarios and rural sections. Adhering to transport regulations and guidelines can mitigate risks associated with hazardous materials. Communication and timely updates about route conditions can further reduce potential hazards for such vehicles.

Truck drivers should be trained to handle varied road conditions, equipped with necessary emergency contact numbers, and aware of alternate routes and local customs for handling hazardous materials. Regular updates on weather and traffic conditions will be crucial for minimizing delays and enhancing safety.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Roundabout	High	11.38395, 77.89480	15 KM/Hr
1	U-Turn	High	10.9400073, 77.43922649999999	10 KM/Hr
2	Turn	High	11.43968, 77.87345	15 KM/Hr
3	Turn	High	11.44022, 77.87546	15 KM/Hr
4	Turn	Medium	11.39976, 77.88382	30 KM/Hr
5	Turn	Medium	11.39690, 77.88852	30 KM/Hr
6	Turn	High	11.37868, 77.89498	15 KM/Hr
7	Turn	Medium	11.37857, 77.89473	30 KM/Hr
8	Turn	High	11.36616, 77.88885	15 KM/Hr
9	Turn	Medium	11.34067, 77.86347	30 KM/Hr
10	Turn	Medium	11.33854, 77.86216	30 KM/Hr
11	Turn	Medium	11.33778, 77.86236	30 KM/Hr
12	Turn	Medium	11.30952, 77.77587	30 KM/Hr
13	Turn	Medium	11.30973, 77.76712	30 KM/Hr
14	Turn	High	11.29490, 77.74685	15 KM/Hr
15	Turn	High	11.29141, 77.74662	15 KM/Hr
16	Turn	Medium	11.28844, 77.74319	30 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
17	Turn	Medium	11.28573, 77.74281	30 KM/Hr
18	Turn	High	11.27319, 77.72981	15 KM/Hr
19	Blind Spot	Blind Spot	11.27330, 77.72715	10 KM/Hr
20	Turn	Medium	11.23054, 77.71796	30 KM/Hr
21	Turn	Medium	11.22387, 77.71385	30 KM/Hr
22	Turn	Medium	11.20776, 77.70916	30 KM/Hr
23	Turn	Medium	11.15942, 77.70068	30 KM/Hr
24	Turn	Medium	11.12102, 77.67919	30 KM/Hr
25	Turn	Medium	11.11932, 77.67786	30 KM/Hr
26	Turn	Medium	11.11710, 77.67175	30 KM/Hr
27	Turn	Medium	11.09491, 77.66603	30 KM/Hr
28	Turn	Medium	11.08296, 77.65857	30 KM/Hr
29	Turn	Medium	11.06183, 77.63924	30 KM/Hr
30	Turn	High	11.00826, 77.56962	15 KM/Hr
31	Blind Spot	Blind Spot	11.00058, 77.56882	10 KM/Hr
32	Turn	High	11.00172, 77.56268	15 KM/Hr
33	Turn	Medium	11.00170, 77.56267	30 KM/Hr
34	Turn	Medium	11.00168, 77.56264	30 KM/Hr
35	Turn	High	10.97018, 77.42713	15 KM/Hr
36	Turn	Medium	10.94222, 77.43831	30 KM/Hr
37	Blind Spot	Blind Spot	10.94001, 77.43923	10 KM/Hr
38	Turn	Medium	10.94014, 77.43907	30 KM/Hr
39	Turn	Medium	10.94240, 77.43809	30 KM/Hr

Emergency Locations

	type	name	coordinates	speed_limit	risk_level
0	hospital	Tiruchengode, Goverment Hospital	11.3903328, 77.8920627	30 km/h	Medium
1	hospital	SPM Medical Centre,Tiruchengode	11.3881331, 77.8931963	30 km/h	Medium
5	clinic	Kongu Nursing Home	11.3783065, 77.8961134	30 km/h	Medium
6	hospital	T.C.A Hospital Tiruchengode	11.3791885, 77.8965774	30 km/h	Medium
7	hospital	Soorya Multispecialty Hospital	11.3786429, 77.8931912	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
8	hospital	Tiruchengode Government Hospital	11.37645, 77.89426	30 km/h	Medium
9	hospital	Tirukumaran Hospitals	11.3782118, 77.8914933	30 km/h	Medium
10	hospital	Krishna Hospital, Namakkal	11.3754811, 77.8931817	30 km/h	Medium
13	hospital	Government Hospital	11.2343326, 77.719226	30 km/h	Medium
16	police	Kangeyam Police Station	11.0011411, 77.562688	30 km/h	Medium
17	hospital	J K Hospital	11.0042298, 77.5603345	30 km/h	Medium
18	hospital	Government Hospital, Koduvai	10.9421891, 77.438567	30 km/h	Medium

Crowded Spots

	type	name	coordinates	speed_limit	risk_level
2	school	அரசு ஆண்கள் மேல்நிலைப் பள்ளி	11.3850439, 77.8948279	30 km/h	Medium
3	school	அரசு பெண்கள் மேல்நிலைப் பள்ளி	11.3844247, 77.8949053	30 km/h	Medium
4	marketplace	திருச்செங்கோடு தினசரி காய்கறி சந்தை	11.3833608, 77.8970145	30 km/h	Medium
11	school	KSR Educational institution	11.3772013, 77.8908807	30 km/h	Medium
12	school	MDV School	11.3719843, 77.8915244	30 km/h	Medium
14	college	Government Arts and Science College, Kangeyam	11.0626738, 77.636227	30 km/h	Medium
15	school	Carmel Girls Higher Secondary School	11.0009019, 77.5710182	30 km/h	Medium

Route Photos of Risky Spots



Risk Type: Roundabout

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.38395, 77.89480



Risk Type: U-Turn

Risk Level: High

Speed Limit: 10 KM/Hr

Coordinates: 10.9400073, 77.43922649999999



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.44022, 77.87546



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.39976, 77.88382



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.39690, 77.88852



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.37868, 77.89498



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.37857, 77.89473



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.36616, 77.88885



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.34067, 77.86347



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.33854, 77.86216



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.33778, 77.86236



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.30952, 77.77587



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.30973, 77.76712



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.29490, 77.74685



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.29141, 77.74662



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.28844, 77.74319



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.28573, 77.74281



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.27319, 77.72981



Risk Type: Blind Spot

Risk Level: Blind Spot

Speed Limit: 10 KM/Hr

Coordinates: 11.27330, 77.72715



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.23054, 77.71796



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.22387, 77.71385



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.20776, 77.70916



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.15942, 77.70068



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.12102, 77.67919



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.11932, 77.67786



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.11710, 77.67175



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.09491, 77.66603



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.08296, 77.65857



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.06183, 77.63924



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.00826, 77.56962



Risk Type: Blind Spot

Risk Level: Blind Spot

Speed Limit: 10 KM/Hr

Coordinates: 11.00058, 77.56882



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.00172, 77.56268



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.00170, 77.56267



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.00168, 77.56264



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 10.97018, 77.42713



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 10.94222, 77.43831



Risk Type: Blind Spot

Risk Level: Blind Spot

Speed Limit: 10 KM/Hr

Coordinates: 10.94001, 77.43923



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 10.94014, 77.43907



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 10.94240, 77.43809
